C++ Fundamentals - Part 3

1. Primitive Types
2. Common STL Types
3. Simple Programmer-Defined Types

Primitive or Fundamental Types (on Windows)

Integral Types - can be signed and unsigned

- char (1 byte)
- short (2 bytes)
- int (4 bytes)
- long (4 bytes)

Floating Point Types

- float (4 bytes)
- double (8 bytes)
- long double (8 bytes)

Boolean (1 byte)

Pointers (4 bytes on 32-bit systems, 8 bytes on 64-bit systems)

See the types and their ranges here:
https://docs.microsoft.com/en-us/cpp/cpp/fundamental-types-cpp

Common STL Types

- Prefer the `string` class over c-style strings

- Prefer `vector`'s over arrays

Simple Programmer-Defined Types

What could go wrong when using this function?
How can we improve this function's signature?

```
const int input = 1;
const int output = 2;
const int append = 3;

// a file can be opened in one of three states
bool open_file(string filename, int open_mode);
```

Scoped Enumeration Types

There is no way to constrain the value being passed into the function. Rather, get the compiler to help enforce our intention:

```
enum class OpenModes { input, output, append };

bool open_file(string filename, OpenModes mode);
```

Usage:

```
open_file("myfile.txt", OpenModes::input);
```

- Enumeration types are useful for modelling small sets of distinct values like *Colour*, *Month* etc.
- Can be used in `switch` and `if` statements